

The Lost Phoenician Artifact: Passive Haptics and Adaptive Movement Gain in Virtual Reality

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This project explores the use of passive haptics and movement gain for Virtual Reality interaction in constrained physical spaces. Passive haptics allows a real physical object, such as a board or table, to act as a tactile proxy for virtual objects. Movement gain remaps the user's real hand movement so that a small physical movement can correspond to a larger, farther, tilted, or differently scaled virtual movement. Building on a previous IGR205 prototype, we reproduced the basic interaction pipeline and improved it with a cleaner calibration workflow, same-hand pinch confirmation, pre-pinch fingertip capture, smooth gain transitions, configurable virtual surface size, configurable tilt, and fingertip-anchored rigid-hand remapping. We also designed a Lebanese-themed VR puzzle game, *The Lost Phoenician Artifact*, where a single calibrated physical board is reused as different virtual objects across several scenes. The current implementation includes a start/calibration/testing scene and a Byblos artifact cabinet, in which movement gain adapts to the depth of the targeted object. The project demonstrates how one passive haptic proxy can support flexible VR interaction through adaptive remapping.

1 Introduction

Virtual Reality can create visually immersive environments, but physical interaction is still limited by the real space around the user. In many VR applications, the user can see virtual objects that appear far away, large, tilted, or placed at different depths. However, the real hand cannot always reach these virtual objects, especially when the user is seated or located in a constrained physical space.

This problem appears in many realistic situations, such as small rooms, dorm rooms, airplanes, trains, or seated setups. In these environments, the user cannot freely walk or extend the arm without hitting real-world boundaries. This creates a gap between the virtual world and the real world: the virtual scene may suggest a large interaction space, while the physical space remains small.

A second problem is the lack of tactile feedback. When a user touches an object in VR, they often touch empty air. This reduces realism, accuracy, and immersion. Passive haptics addresses this issue by using real physical objects as tactile proxies for virtual objects. Movement gain remaps the user's real hand movement so that a small physical movement becomes a larger or differently positioned virtual movement. For example, with a gain of 2, moving the real hand by 10 cm can move the virtual hand by approximately 20 cm.

Our project is implemented as a Lebanese-themed VR puzzle game called *The Lost Phoenician Artifact*. The player explores scenes inspired by Lebanese heritage, showcasing many concepts of VR.

2 Previous Work and Motivation

2.1 Previous IGR205 Project

The previous IGR205 project, *Combining Passive Haptics and Movement Gain in Virtual Reality*, implemented a first prototype combining passive haptics and movement gain. Their system calibrated one real physical surface and used it as a haptic proxy for virtual

interaction. The user could touch the real surface while seeing a virtual surface in the VR scene.

Their prototype had three main goals:

- (1) calibrate a physical object to match a virtual object;
- (2) apply movement gain to amplify the user's hand movement;
- (3) dynamically adjust gain according to the virtual interaction.

2.2 Limitations of the Previous Work

The previous project showed that passive haptics and movement gain can work together, but several limitations remained.

First, the calibration workflow was difficult to use. The user had to touch a point with one hand and press a validation button with the other hand, which was far and hard to reach in some cases. This two-hand workflow was technically functional, but it was not natural or comfortable.

Second, the calibration implementation was not clean enough for reuse in a larger project. The system used a general corner interpretation method, and during our own reproduction, we observed that this could create confusion about which point corresponds to which corner. A wrong interpretation of the points could make the board appear with the wrong orientation.

Third, the gain system was mainly palm-based. This is understandable because moving the palm moves the whole hand without deforming it. However, the actual contact point during surface interaction is the index fingertip, not the palm. This can create a mismatch where the palm is correctly remapped, but the fingertip does not exactly reach the visible virtual surface.

Finally, the previous work did not focus on adaptive movement gain for individual 3D objects. The gain was mainly connected to the haptic surface and its direction. In our project, this became a problem when we started placing small objects deeper along the virtual depth axis. If an object is small and far in the virtual z direction, a fixed surface gain is not enough. The system needs to adapt the gain according to the specific object that the user is trying to reach.

2.3 Our Objective

Our objective was not only to reuse the previous project, but to improve it. We first reproduced the base system: OpenXR hand tracking, three-point calibration, virtual surface alignment, and movement gain. Then we focused on solving the main interaction problems.

Our improvements are:

- cleaner three-point calibration using a fixed point order;
- same-hand pinch confirmation instead of two-hand button validation;
- pre-pinch fingertip capture to avoid recording shifted pinch positions;
- saving calibration scale for later scenes;
- smooth gain transitions;

- configurable board size;
- configurable board tilt;
- fingertip-anchored rigid hand remapping;
- VR testing panel for gain, size, and tilt;
- adaptive object-depth gain.

3 Game Design and HCI Considerations

3.1 Narrative and Creative Direction

Rather than creating a simple technical demonstration, we wanted to embed our interaction framework into a meaningful experience. This led to the creation of *The Lost Phoenician Artifact*, a VR puzzle-adventure game inspired by Phoenician history and Lebanese cultural heritage. The player explores locations such as Byblos, the Cedars of Lebanon, and Baalbek while solving puzzles to uncover a legendary lost artifact. This theme provided both a coherent narrative structure and an opportunity to showcase elements of Lebanese history and culture.

3.2 HCI Considerations

Before implementing the game, we spent time creating sketches and discussing different storylines, puzzle ideas, and interaction layouts. These sketches helped us visualize how the narrative would progress across the different locations and how the physical board could be reused in meaningful ways throughout the adventure. This design process allowed us to connect the technical aspects of passive haptics with a coherent gameplay experience.

Several HCI principles guided our design choices. To reduce cognitive load, the user always interacts with the same physical board even though its virtual appearance changes throughout the game. We also introduced visual feedback such as highlighting and hover effects to clearly indicate which object is being targeted. A dedicated calibration and testing scene was added to familiarize users with the interaction techniques before gameplay. Finally, smooth gain transitions were implemented to avoid abrupt changes in virtual reach and provide a more comfortable interaction experience.

To demonstrate our system, we designed a Lebanese-themed VR puzzle game called *The Lost Phoenician Artifact*. The player explores locations inspired by Lebanese heritage while searching for a legendary lost Phoenician artifact. Throughout the game, the same physical board is reused as different virtual objects, demonstrating the flexibility of passive haptics and movement gain.

4 Technical Details

4.1 Unity and XR Setup

The project is implemented in Unity using OpenXR and XR Hands. We use OpenXR because movement gain requires direct control over the virtual hand transforms. A standard interaction setup is not enough, because the virtual hand must sometimes move differently from the real tracked hand.

The base scene contains: XR Origin; hand tracking system; left and right hand visualizers; calibration manager; gain manager; virtual surface object; calibration markers; VR control panel; testing panel.

4.2 Calibration System

The calibration system records three points on the real board. We use a fixed explicit order:

- (1) top-left;
- (2) top-right;
- (3) bottom-left.

This fixed order is important because it avoids fragile corner-sorting logic. The user knows exactly which corner to touch, and the system can compute the surface orientation directly.

From the three points, the system computes:

- visual top-left, top-right, bottom-left, bottom-right points;
- surface center, normal, rotation, width, height, and scale.

The virtual surface is then placed at the computed center, rotated according to the computed surface rotation, and scaled according to the calibrated width and height.

We also save the calibration data so that later scenes can reuse it. This includes the position, rotation, normal, and scale. Saving the scale was necessary because, without it, later scenes could load the surface with the wrong dimensions.

4.3 Same-Hand Pinch Confirmation

The previous project used a two-hand validation method. We replaced this with same-hand pinch confirmation. The user touches the corner with the index fingertip and confirms the point by pinching with the same hand.

To solve this, we implemented pre-pinch fingertip capture. While the hand is open, the system continuously stores the last stable index fingertip position. When a pinch is detected, the system records the stored pre-pinch position instead of the shifted pinch position.

4.4 Gain Manager

The gain manager controls the remapping between the real hand and the virtual hand. The basic idea is:

$$d_{\text{virtual}} = g \cdot d_{\text{real}}$$

where g is the gain value. If $g = 1$, the virtual movement matches the real movement. If $g > 1$, the virtual hand moves farther than the real hand. transform from the transformed virtual transform. This avoids unstable behavior when changing gain, size, or tilt.

4.5 Smooth Gain Transitions

A direct jump from gain 1 to gain 2 or gain 3 can be confusing. The virtual hand and surface suddenly move, and the user may lose the connection between real and virtual movement.

To solve this, we added smooth gain transitions. Instead of changing instantly, the gain interpolates from the current value to the target value over a short duration. This makes the movement easier to follow.

4.6 Configurable Board Size and Tilt

The virtual board can be resized using a board size multiplier. This allows the same physical board to visually become a larger or smaller virtual object.

The main technical requirement is that resizing and tilting should not only move the visual board. They must also affect the hand

remapping. If the user touches a real corner, the virtual fingertip should touch the corresponding transformed virtual corner.

4.7 Fingertip-Anchored Rigid Hand Remapping

The previous implementation mainly used the palm as the reference for gain. This is useful because the palm is a stable parent for the hand, but it is not the actual contact point. During surface interaction, the important point is the index fingertip.

Our solution is fingertip-anchored rigid hand remapping. The system:

- (1) reads the real index fingertip position;
- (2) maps this fingertip to the corresponding point on the transformed virtual surface;
- (3) computes the offset between the real fingertip and the desired virtual fingertip;
- (4) applies this offset rigidly to the whole virtual hand.

4.8 Visual Design and Assets

We also paid attention to the visual design of the environments.

During development, we searched for assets that closely matched Lebanese historical and cultural settings. However, many of the most suitable asset packs were commercial and outside the scope of our project budget. As a result, we selected freely available Unity assets and combined them to create environments that captured the desired atmosphere while remaining visually coherent.

The scenes were manually assembled in Unity, including the placement of buildings, decorations, props, lighting, and interaction elements.



Fig. 1. Cedar Forest Scene

5 Game Scenes

5.1 Scene 0: Calibration Scene

The calibration scene serves as the entry point of the application. Users calibrate the physical board using the same-hand pinch workflow. This scene demonstrates the improved calibration process and ensures that the virtual surface remains aligned with the real board throughout the experience.

5.2 Scene 1: Byblos Game 2: Adaptive Gain Game

The Adaptive Gain Game showcases adaptive object-depth gain. The player must select objects in the correct sequence while interacting with artifacts positioned at near, medium, and far depths. The system



Fig. 2. Calibration scene used to align the physical board with its virtual representation before gameplay.

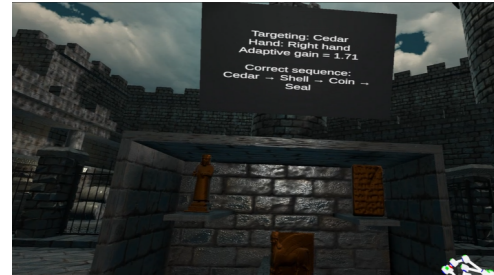


Fig. 3. Adaptive Gain Game in the Byblos scene. Objects are placed at different virtual depths.

automatically adjusts the gain according to the depth of the targeted object, allowing distant objects to be reached while using the same physical board.

5.3 Scene 1: Byblos Game 2: Maze Game



Fig. 4. Maze game based on continuous drag interaction.

The Maze Game demonstrates continuous surface interaction. The player navigates through a maze by dragging their fingertip along a valid path displayed on the virtual board. This scene highlights precise fingertip tracking and the reuse of the same passive haptic surface for a different gameplay mechanic.

5.4 Scene 2: Cedars Forest Game 1: Memory Game

The Memory Game was designed to combine cognitive tasks with passive haptic interaction. The player observes and memorizes a sequence before reproducing it correctly. During development, both 2D and 3D versions of the game were explored to evaluate different interaction approaches. We had some difficulties in making the 3D version work, for the project, we kept the 2D version.



Fig. 5. Memory game prototype developed for the Cedars Forest scene.

5.5 Scene 2: Cedars Forest Game 2: Fitts' Law Game



Fig. 6. Fitts' Law inspired game used to test board transformations and interaction performance.

The Fitts' Law Game : It showcases board transformations, particularly the tilt feature. Although the physical board remains flat, the virtual board can appear at different angles. The remapping system compensates for these transformations to maintain accurate fingertip contact.

6 Team Contributions

6.1 Karen El Khoury

- Developed the Cedars Forest scene.
- Implemented the Fitts' Law game.
- Implemented board tilt transformations.
- Contributed to sketches, report writing, and presentation preparation.

6.2 Mohamad Hassan Ibrahim

- Developed the Byblos scene.
- Implemented the Maze Game.
- Implemented the fixed gain system.
- Contributed to sketches, report writing, and presentation preparation.

6.3 Hadi Jaber

- Improved and fixed the calibration system.
- Developed the Calibration Scene.
- Implemented the Adaptive Gain Game.
- Explored both 2D and 3D Memory Game prototypes.
- Contributed to sketches, report writing, and presentation preparation.

7 Observations & Future Work

During development, we observed that the main difficulty is not only applying gain mathematically. The real challenge is maintaining a believable relationship between real touch and virtual contact. If the board is moved, resized, tilted, or used to represent deeper objects, the virtual fingertip must still appear to touch the correct point.

We also observed that calibration usability is very important. A calibration method can be technically correct but still feel bad if it requires too many steps or unnatural hand coordination. Same-hand pinch makes calibration easier, but it required pre-pinch capture to remain precise.

The adaptive Byblos cabinet also showed that object depth matters. When objects are small and deep in the virtual scene, fixed gain is not enough. The gain must adapt based on the intended target.

Future work includes:

- completing the visual design of the Byblos artifact cabinet;
- replacing primitive artifacts with better 3D models;
- tuning near, medium, and far depth values;
- adding sound effects for hover, correct, wrong, and completion feedback;
- completing the Cedars map puzzle;
- completing the Baalbek final puzzle;
- adding user testing;
- logging task completion time, gain values, errors, and target depth;
- comparing adaptive gain with fixed gain;
- adding automatic real surface detection;
- supporting multiple physical surfaces.

8 Conclusion

In this project, we explored how passive haptics and movement gain can be combined to improve VR interaction in constrained physical spaces. Starting from a previous IGR205 prototype, we reproduced the core interaction pipeline and introduced several improvements, including a cleaner calibration workflow, same-hand pinch confirmation, configurable board transformations, smooth gain transitions, fingertip-anchored remapping, and adaptive object-depth gain.

To demonstrate these techniques, we developed *The Lost Phoenician Artifact*, a Lebanese-themed VR puzzle game in which a single physical board is reused as multiple virtual objects across different scenes. Through the Calibration Scene, Adaptive Gain Game, Maze Game, Memory Game, and Fitts' Law Game, we showed how one passive haptic proxy can support a variety of interaction techniques while maintaining consistency and immersion.

References

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